Maintenance Schedule

Diesel engine
12V1600Gx1F
12V1600Gx1S
Application group 3D

MS50811/02E

A Rolls-Royce solution

Engine model	kW/cyl.	Application group
12V1600G71F	61.7 kW/cyl.	3D, Emergency Standby power, IFN, 50 Hz
12V1600G81F	65.6kW/cyl.	3D, Emergency Standby power, IFN, 50 Hz
12V1600G91F	74 kW/cyl.	3D, Emergency Standby power, IFN, 50 Hz
12V1600G51S	60.3 kW/cyl.	3D, Emergency Standby power, IFN, 60 Hz
12V1600G61S	65.3 kW/cyl.	3D, Emergency Standby power, IFN, 60 Hz
12V1600G71S	69.7 kW/cyl.	3D, Emergency Standby power, IFN, 60 Hz
12V1600G81S	74.2 kW/cyl.	3D, Emergency Standby power, IFN, 60 Hz
12V1600G91S	83 kW/cyl.	3D, Emergency Standby power, IFN, 60 Hz

Table 1: Applicability

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1 Maintenance Schedule

1.1 Preface

This manual provides instructions for your product manufactured by Rolls-Royce Solutions.

Maintenance concept

The maintenance system for Rolls-Royce Solutions products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of operational availability. The maintenance intervals as well as the required maintenance task scopes are based on operational experience and are therefore to be considered as recommendations. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of demanding operational and ambient conditions.

Observance of the specified maintenance intervals is essential to maintain product safety.

The intervals at which the maintenance tasks are to be performed are specified as operating hours and time limits. Whichever value is reached first shall apply.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of Rolls-Royce Solutions. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product
QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule)
QL3: Maintenance work which requires partial disassembly of the product
QL4: Maintenance work which requires complete disassembly of the product

Additional notes on maintenance and preservation:

Maintenance activities stipulated in any documentation provided by third-parties (suppliers and component manufacturers), particularly those relating to individual components and subsystems, may differ from the information specified in this Maintenance Schedule. The Maintenance Schedule provided by Rolls-Royce Solutions shall take precedence in such case.

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

Refer to the applicable Fluids and Lubricants Specifications for details of fluid and lubricant change intervals.

If the product is to remain out of service for a longer period of time, Rolls-Royce Solutions recommends carrying out a monthly test run until the engine has reached its normal operating temperature. If the engine is scheduled to remain out of service for more than 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications.

The latest version of the specifications is available at http://www.mtu-solutions.com.

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1.2 Special conditions

Designation	Value
Load factor	≤ 85%
Operating hours	500 h per year or for the duration of an emergency
Time between major overhauls	8,000 h or 18 years

Table 2: Special conditions

Note 01:
The engines are certified for operation with approved fuels as listed in the Fluids and Lubricants Specifications. The component maintenance intervals specified in the maintenance schedule refer to engine operation with diesel fuel with a sulfur content < 15 ppm, e.g. EN 590. If fuels with a sulfur content > 15 ppm are used, the maintenance intervals specified in the maintenance schedule for the affected components could be shorter. Engines with exhaust gas recirculation and/or exhaust gas aftertreatment must not be operated with excessive sulfur content in the fuel. The limit values in the Fluids and Lubricants Specifications apply.

Table 3: Special conditions

Note 02:
If the engine is used in an UPS system with corresponding hardware, the exchange interval for the cylinder heads is 500 h.

Table 4: Special conditions

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